Market Making on Binance Perpetual Futures: Testing the Avellaneda–Stoikov Model and Measuring the Costs and Returns of Liquidity Provision

Project Proposal
Statistical Consulting - STA 334
Fall 2026

Muhammad Abdullah
mabdullah@conncoll.edu

Due: October 16th 2026

Abstract

Market makers quote prices to buy and sell continuously and earn the gap between them, while risking losses to price moves and to better-informed traders. This project studies that trade-off on 90 days (1 January to 30 March 2024) of best-bid/ask quotes and trades for three Binance perpetual futures contracts that span a range of liquidity: ETHUSDT, LINKUSDT and MATICUSDT. It has two parts. The first tests the three core assumptions of the Avellaneda–Stoikov (A–S) market-making model, which underlies many strategy simulators (constant volatility, exponentially decaying fill probability, and no information in order flow). The second measures what matters to people who trade (what market makers actually earn after adverse selection and fees, how trading costs grow with order size, when liquidity is cheapest, and whether ETH price moves lead the smaller coins). Each question is answered with a named statistical model (binomial GLMs, logistic and mixed-effects regression, vector autoregression, survival analysis) and reported in a way that a client can act on (ticks, basis points, seconds and dollars). The project and it's findings can statistically consult two types of clients: crypto market-making desks (companies like Wintermute, Keyrock, JumpCrypto etc), and research groups that build market-making simulators (e.g. Al Autonomous learning Lab - AALL at Connecticut College lead by Professor Gary Parker and Jim O'Connor)

1 Introduction

1.1 Background

On an exchange, buyers post *bids* (offers to buy) and sellers post *asks* (offers to sell). The highest bid and lowest ask are the *best bid* and *best ask*; their midpoint is the *mid price* and the gap between them is the *spread*. A *market maker* posts a bid and an ask at the same time and earns the spread when both sides trade. Two risks offset that income. The first is that the price moves while the market maker holds inventory. The second is *adverse selection*: the traders most eager to trade with a market maker are often the ones who know, or guess correctly, that the price is about to move against it. On top of both, the exchange charges a fee on every trade.

Avellaneda and Stoikov (2008) gave the best-known mathematical treatment of this problem. Their model tells a market maker where to place quotes given its inventory, its risk aversion, the price volatility, and how often quotes at different distances get filled. It has since been extended (Guéant et al., 2013) and is a standard teaching model (Cartea et al., 2015). It also underlies simulation-based research: the EvoMM project, (on which I am the Co-author and presented in Tokyo, Japan in September (O'Connor et al., 2026)), applies quality-diversity search to market-making strategies inside an A–S simulator. A strategy that performs well in such a simulator is only as trustworthy as the simulator's assumptions.

1.2 Clients

The project serves two clients with equal priority.

1. **A crypto market-making desk.** Its decisions like which coins to quote, how far from the mid to quote, when to widen or step back, how much it actually keeps per trade after adverse selection and fees, and whether it should adjust quotes on smaller coins when ETH moves. The desk can be companies like Wintermute, Keyrock, JumpCrypto etc.
2. **Research groups that build market-making simulators**, such as the group led by Prof Gary Paker and Prof. Jim O'Connor that developed the EvoMM simulator. Their decision: when they extend a simulator, which A–S assumptions are most wrong in practice, and what realistic parameter values should replace the textbook ones? The code from this project is also meant to be reused for their future projects.

Two secondary audiences benefit from the same results without driving the questions - an **execution trader** at a fund who needs to buy or sell size at low cost, and an **active individual trader** deciding when to trade and whether a limit order is worth it.

1.3 Why three coins

ETHUSDT, LINKUSDT and MATICUSDT form a ladder from very liquid to much less liquid, observed on the same exchange over the same days. That turns a single-asset case study into a comparison, which is what practitioners use: "how much more does it cost to trade MATIC than ETH?" and "where can a small market maker compete?" cannot be answered from one coin. Because all three spreads turn out to be pinned at one tick, the coins differ mainly in how much a tick is worth relative to the price (from a few hundredths of a basis point for ETH to over one basis point for MATIC). That makes the comparison a natural test of how tick size shapes what market makers earn. It also helps the A–S tests, because a parameter that changes sharply across coins cannot be treated as a universal constant in a simulator. (MATIC was later renamed POL; the data use the ticker in force in early 2024.)

1.4 Research questions

The questions fall into two parts.

Part A: The economics of liquidity (core questions)

Q1 What do market makers earn? How does the cost paid by traders (the effective spread) split into what the market maker keeps (the realized spread) and what it loses to subsequent price moves (price impact), by coin, trade size, and time of day?

Q2 Can a small market maker cover fees? At realistic fee levels, what is the probability that a passive fill is profitable, and on which coins is market making viable for a participant who does not receive the lowest fees?

Part A: Secondary questions

Q3 Liquidity calendar: When is each coin cheapest and most expensive to trade, by hour of day and day of week?

Q4 Lead-lag: Do ETH price moves predict LINK and MATIC price moves over the next few seconds, and how much of an ETH move carries over?

Part B: Testing the Avellaneda-Stoikov assumptions

Q5 Volatility: Is mid-price volatility constant, or does it cluster and follow an intraday pattern? How large and how persistent are the departures, and do they make A-S quotes materially wrong? *(A-S assumption 1)*

Q6 Return distribution: How close to normal are mid-price returns at 1 second, 10 seconds, and 1 minute? Are the heavy tails explained by changing volatility, or are there genuine jumps? *(Assumption 1)*

Q7 Fill intensity: Does the probability that a quote is reached fall off exponentially with its distance δ from the mid as $\lambda(\delta) = Ae^{-k\delta}$? What are A and k for each coin, and does a different shape fit better? *(Assumption 2)*

Q8 Adverse selection: After a buyer- or seller-initiated trade, how far does the mid move in the aggressor's direction over the next 1–60 seconds, and how much does recent order flow predict the next price move? *(Assumption 3)*

2 Data

2.1 Source and coverage

All data come from Binance's free public archive (Binance, 2024) for three USD-M perpetual futures contracts: ETHUSDT, LINKUSDT and MATICUSDT. For each coin, two daily file types are used:

- **bookTicker (quotes).** One row each time the best bid or best ask changes: best bid price and size, best ask price and size, an update ID, the matching-engine time (`transaction_time`) and the publication time (`event_time`), both in milliseconds UTC.

Binance stopped publishing futures bookTicker files in 2024, so quote data exist only for a fixed historical window. After download, longest unbroken run of days with complete files for all three

coins is **1 January to 30 March 2024 (90 days)**: 6 files per day (two file types for each of three coins), 540 files in total. This window is the study sample.

The sample contains two well-known market events worth flagging in advance: the U.S. approval of spot Bitcoin ETFs on 10 January 2024, and Ethereum's Dencun upgrade on 13 March 2024. Neither is a research question, but both are likely to show up as unusual days and will be identified in the data rather than excluded by assumption.

2.2 Scale

A single ETHUSDT pilot day (5 October 2023) contained about 7.0 million quote updates and 1.9 million trades. LINK and MATIC are less active, so ETH will dominate the data volume: several hundred million quote rows over 90 days. Raw files are converted once to Parquet, sorted and typed, and queried with DuckDB so the full sample never has to sit in memory. Most analysis uses derived tables (one row per second or per minute per coin, and one row per aggregated trade), which are small enough for ordinary Python tools on my laptop.

2.3 Known data issues

- **Row order.** Users of the archive reported out-of-order rows in ETH and BTC bookTicker files from January 2024 onward, which is the start of this sample. All files are re-sorted by time and update ID before any time-based calculation, and the sort is verified.
- **Tick size differs by coin.** The smallest price step is a much larger fraction of the price for MATIC (trading near $1) than for ETH. Preliminary data show that all three spreads sit at one tick almost all the time (Section 2.4), so the cost of a single tick, not competition between quotes, sets the spread on every coin.
- **Top of book only.** The data show the best bid and ask, not the full order book. Queue position and depth beyond the best price are unobservable.
- **Missing trades.** Trade IDs are sequential, so gaps reveal missing trades. On the pilot day there were 113 ID jumps out of 1.9 million trades; every coin and day will be checked.

2.4 Preliminary exploration

One ETHUSDT pilot day (5 October 2023, outside the study window - used for EDA). The file passed every integrity check: no duplicate update IDs, no crossed or locked quotes, no gaps longer than 60 seconds, and full coverage from 00:00 to 23:59 UTC. The mid price was unchanged in 75% of one-second intervals, so at the shortest horizon the price behaves like a discrete jump process rather than a continuous random walk. Trade sizes were extremely skewed: the median trade was about $50, the mean about $1,600, and the largest about $4.8 million.

All three coins on two January weekdays (24–25 January 2024). These days fall in the training months. The five figures below preview the main questions; every number in this section comes from these two days only and will be re-estimated on the full sample.

- **Every spread is pinned at one tick** (Figure 1). All three coins spent essentially the entire two days at the minimum spread. What differs is how much that one tick is worth -

the time-weighted half-spread, which is what a market maker earns per fill before fees, was roughly 0.02 bps for ETH, 0.36 bps for LINK and 0.69 bps for MATIC. Even the largest is well below the base-tier maker fee of about 2 bps. Liquidity on these coins therefore varies through the size waiting at the best quotes, not through the spread.

- **Fill probability decays roughly exponentially, but not exactly** (Figure 2). On a log scale the LINK and MATIC curves are close to straight lines, which is what the A–S model assumes. The ETH curve bends: quotes far from the mid are reached more often than an exponential predicts (about 2×10^{-4} observed at 20 bps against about 1×10^{-4} fitted). ETH quotes are also reached least often at any given distance, because ETH moves fewer basis points in ten seconds.
- **Market makers lose to adverse selection on every coin** (Figure 3). After a trade, the mid moved further in the taker's direction than the half-spread the maker collected. For ETH the move grew from about 0.33 bps at 1 second to 0.45 bps at 60 seconds, roughly twenty times the half-spread. LINK and MATIC peaked near 0.9 and 0.8 bps at 5 seconds and then partly reversed, which suggests part of their loss is temporary price pressure rather than information. This contradicts the A–S assumption that fills carry no information and motivates Q4–Q6 directly.
- **Heavy tails on all three coins** (Figure 4). Standardized returns reached about 30 standard deviations at 1 second and still about 9 at 1 minute. The coins look alike, which suggests heavy tails are a property of this market rather than of one contract.
- **A shared daily rhythm** (Figure 5). All three coins were quiet from about 02:00 to 09:00 UTC and busiest from about 10:00 to 18:00 UTC, with volatility up to about 1.45 times its median hour and volume up to about twice its median hour, and a low near 22:00 UTC. The coins move together closely, an early hint for the commonality question. With two days, single-hour spikes may be single events, so only the overall shape should be read.

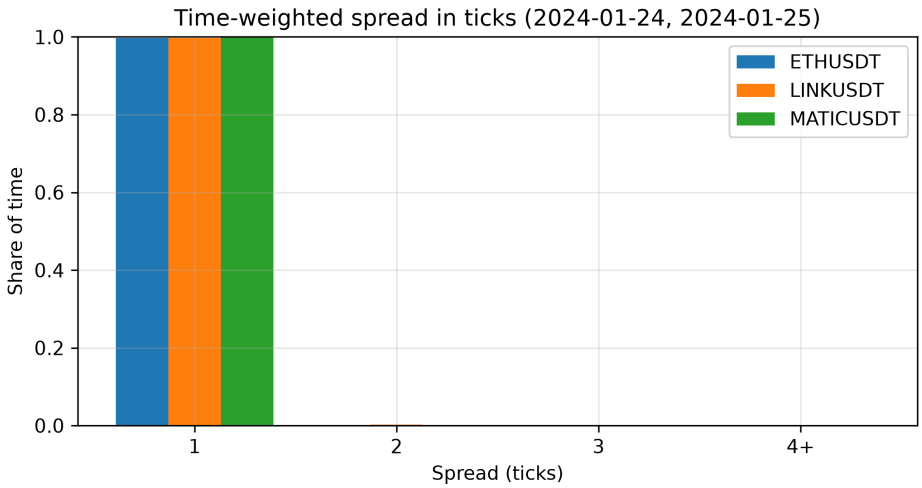


Figure 1: Share of time each coin's spread sat at 1, 2, 3 or 4+ ticks, weighting each quote by how long it was live (24–25 January 2024). All three coins are at one tick essentially all the time.

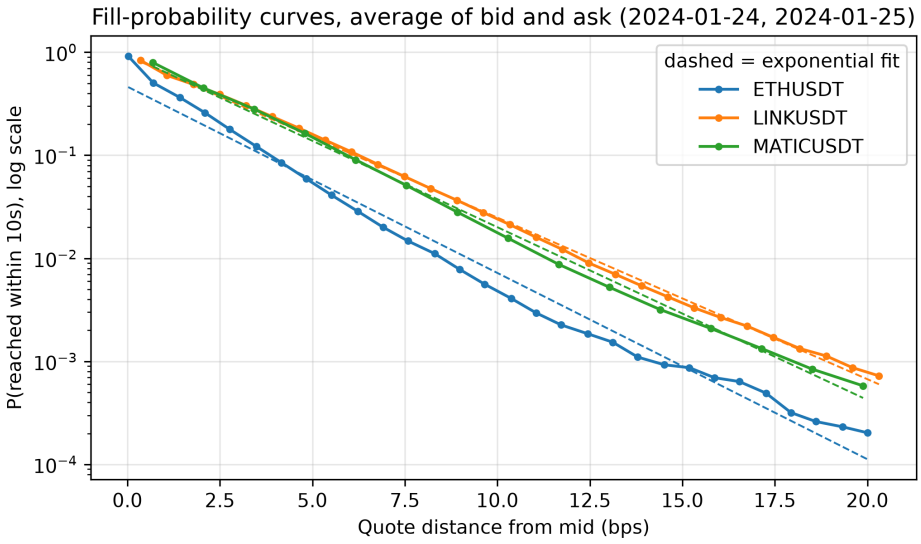


Figure 2: Fill-probability curves (24–25 January 2024). Hypothetical quotes are placed on real price levels behind the best bid and ask at the start of each 10-second window; the curve shows how often a trade reached each level, averaged over both sides. A straight line on the log scale means exponential decay, the A–S assumption; dashed lines are exponential fits. "Reached" is an upper bound on "filled", because a resting order may sit behind others in the queue.

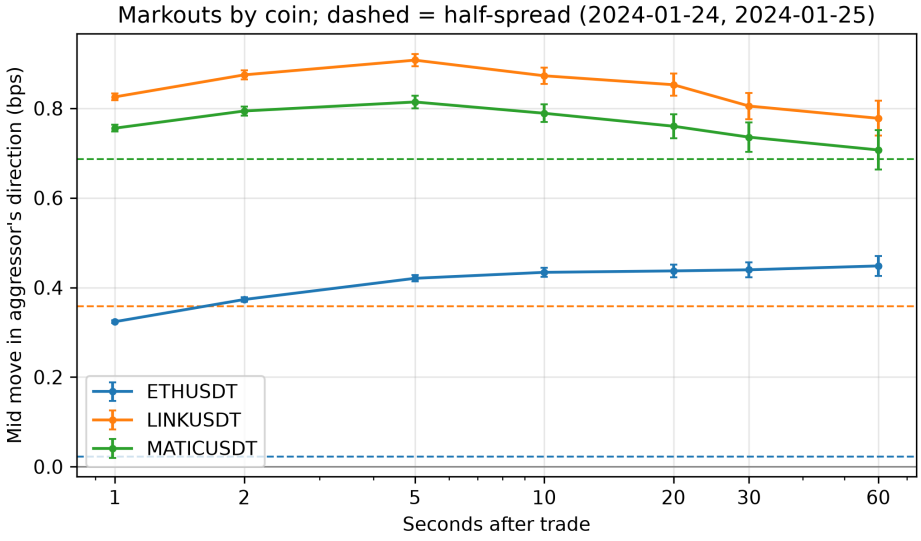


Figure 3: Average mid-price move after a trade, in the direction of the taker, 1–60 seconds later (24–25 January 2024). Dashed lines show each coin's time-weighted half-spread, what a market maker earns per fill before fees. Curves above their dashed line mean the average maker loses money before fees. Error bars ignore dependence between nearby trades and are too narrow; the alignment check has not yet been run.

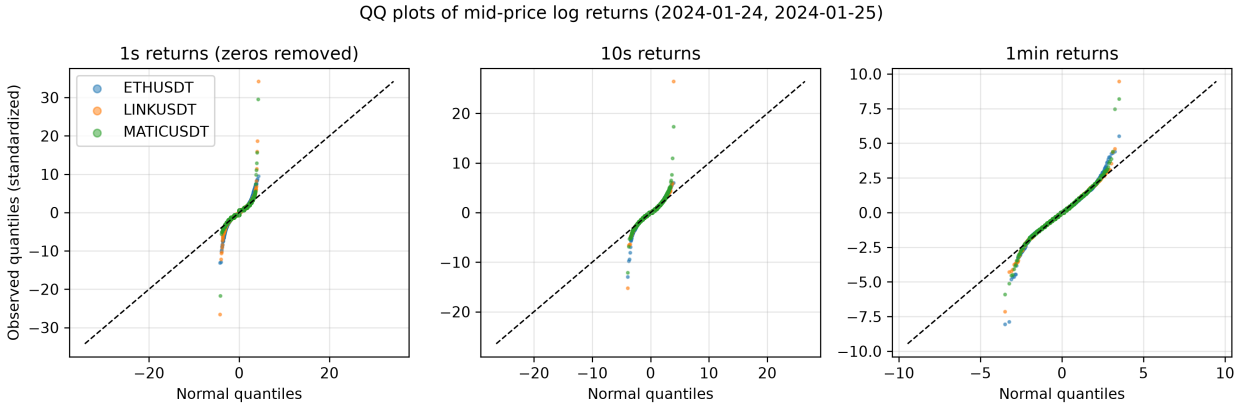


Figure 4: QQ plots of standardized mid-price log returns against the normal distribution at 1 second (zero returns removed), 10 seconds and 1 minute (24–25 January 2024). Points on the dashed line would indicate normality.

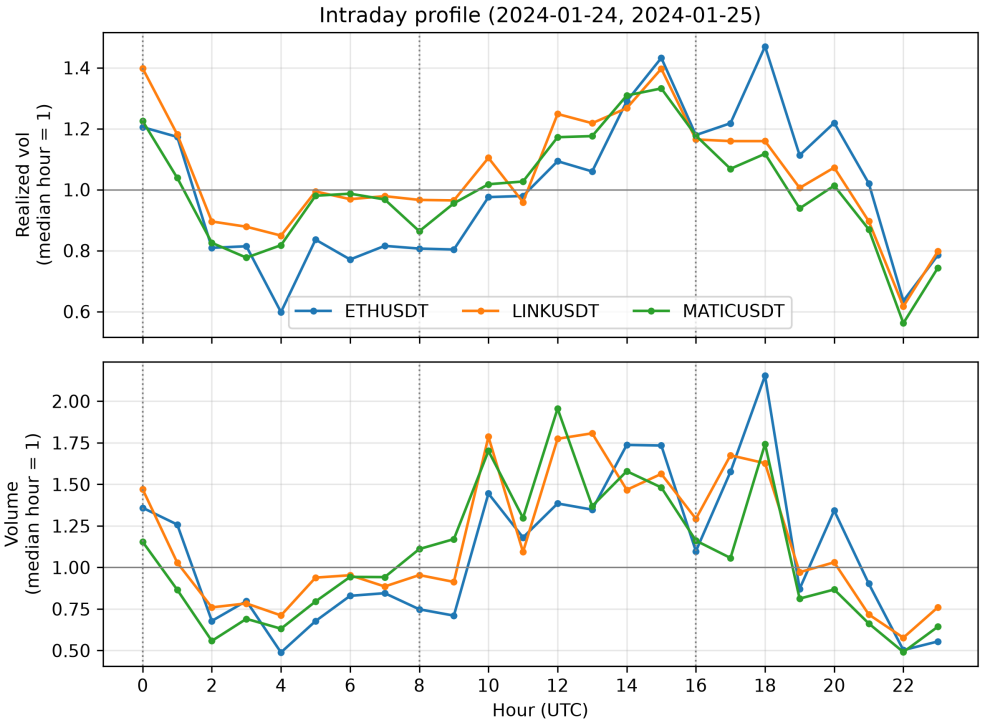


Figure 5: Median one-minute realized volatility and average hourly volume by UTC hour, each scaled so the coin's median hour equals 1 (24–25 January 2024). Dotted lines mark funding times.

3 Methodology

3.1 Data preparation

Every step is run separately for each coin.

1. **Integrity checks**, for every coin and day: time coverage, duplicates, crossed or locked quotes, gaps between quote updates, publication lag (`event_time` minus `transaction_time`), zero-size quotes, and trade-ID gaps. Any gap longer than 60 seconds is treated as missing data and excluded. All findings are logged and reported.
2. **Trade–quote alignment.** Each trade is matched to the last quote strictly before it on the matching-engine clock. If signing and timestamps are correct, buyer-initiated trades should print at or above the best ask and seller-initiated trades at or below the best bid. The share of trades that agree is reported, and a sweep of time offsets (± 500 ms) checks for a systematic lag. A misclassification rate p shrinks measured adverse selection by a factor of $(1 - 2p)$, so agreement of at least 95% is required before markouts are taken at face value; 85–95% is usable with the attenuation reported; below 85% the signing is investigated before going further.
3. **Derived variables.** Mid price $m_t = (b_t + a_t)/2$; spread in ticks and in basis points; quote imbalance $(q_t^b - q_t^a)/(q_t^b + q_t^a)$; trade sign $s = +1$ for buyer-initiated and -1 for seller-initiated; tick size inferred from the data.
4. **Clock-time sampling.** Quotes arrive at irregular times, so the mid is sampled on fixed grids (1 s, 10 s, 1 min) using the last quote in each interval, and returns are log returns $r_t = \ln(m_t/m_{t-1})$, reported in basis points. All three coins share the same grid, which the lead–lag analysis needs.
5. **Time-weighting.** Quote-level averages (spread, quoted size) weight each quote by how long it was live, capped at 60 seconds, so that bursts of rapid updates do not dominate.
6. **Parent trades.** Trades with the same millisecond, the same aggressor side and consecutive IDs are merged into one "parent" trade, since one large order usually fills against several resting orders at once. Results are checked with and without this aggregation.
7. **Train and test split.** January and February 2024 (60 days) are used for exploration and model building. March 2024 (30 days) is held out and used only to evaluate fitted models.

3.2 Part A: The economics of liquidity

Q1: What market makers earn. For a trade at price p, sign s and mid m_t, with the mid m_{t+h} seconds later, the standard decomposition (Huang and Stoll, 1996; Hasbrouck, 2007) is

$$\underbrace{2s\,\frac{p - m_t}{m_t}}_{\text{effective spread}} = \underbrace{2s\,\frac{p - m_{t+h}}{m_t}}_{\text{realized spread}} + \underbrace{2s\,\frac{m_{t+h} - m_t}{m_t}}_{\text{price impact}},$$

all in basis points, for $h \in \{5, 30, 60\}$ seconds. The effective spread is what the taker paid relative to the mid; the realized spread is what the market maker kept after the price moved; price impact is what it lost to adverse selection.

- **Linear mixed-effects regression** (Laird and Ware, 1982) of each component on coin, trade-size bucket, hour of day and trailing volatility, with a random intercept for each day.
- **Quantile regression** (Koenker and Bassett, 1978) at the 50th, 90th and 99th percentiles, because the bad trades matter more than the average one.
- Size buckets are percentiles of parent-trade notional within each coin (0–50, 50–90, 90–99, 99+), fixed on the training months.

Q2: Can a small market maker cover fees? A maker's gain on one fill, before fees, is half the realized spread, $g = s\,(p - m_{t+h})/m_t \times 10^4$ bps.

- Binance's base-tier fees in early 2024 were roughly 2 bps for makers and 5 bps for takers, with lower fees and rebates for high-volume participants. Preliminary data put the half-spread below this fee on all three coins (Figure 1), so base-tier fees likely exceed what a maker earns from the spread everywhere, and the analysis is run across a range of maker fees from a rebate to the base rate rather than at one value.
- **Logistic regression** of $P(g > f)$, the probability that a passive fill is profitable at fee level f, on coin, size bucket, hour and trailing volatility. Evaluated on March data by AUC, Brier score and calibration.
- The output is a break-even table: for each coin, the maker fee at which the average fill stops being profitable. This tells a desk which coins it can compete in at its fee level, and tells an individual trader how much of the cost of a market order is fee versus spread.

Q3: Price impact and size. **Log-log regression** of the absolute mid move after a parent trade on its notional value, with coin interactions and Newey–West errors. The slope is tested against 0.5, the "square-root law" of price impact reported in other markets (Tóth et al., 2011). Binned means by size decile are shown alongside as a model-free check. The output is a rule of thumb an execution trader can use: the expected cost of a \$10k, \$100k or \$1m order in each coin.

Q4: Liquidity calendar. Because spreads are pinned at one tick, liquidity is measured mainly by the time-weighted quantity at the best bid and ask, with the share of time the spread is wider than one tick as a second outcome. **Linear mixed-effects models** of log quoted size, and a **logistic regression** of whether the spread is wider than one tick in a given minute, are fitted on hour-of-day and weekday/weekend indicators interacted with coin, with a random intercept for each day. Quantile regression targets the worst minutes. Results will be shown as heatmaps per coin: the cheapest and most expensive hours to trade.

Q5: Lead–lag between ETH and the smaller coins.

- **Cross-correlation functions** of 1-second returns for each pair, at leads and lags up to 30 seconds.
- A **vector autoregression** (VAR; Sims, 1980) of the three coins' 1-second returns, with lag order chosen by BIC, and **Granger causality tests** (Granger, 1969) in each direction using HAC-robust Wald statistics. Impulse response functions show how much of a one-basis-point ETH move appears in LINK and MATIC, and how quickly.

- Because prices update at irregular times, the Hayashi–Yoshida covariance estimator (Hayashi and Yoshida, 2005), built for asynchronous observations, is used as a robustness check.
- The desk takeaway is how long, in seconds, a LINK or MATIC quote stays stale after an ETH move.

3.3 Part B: Testing the A–S assumptions

Q5: Volatility.

- **Measurement.** Realized volatility in each minute is the square root of the sum of squared 1-second returns. A volatility signature plot (Andersen et al., 2000), with realized variance averaged over all starting offsets (Zhang et al., 2005), sets the shortest sampling interval at which microstructure noise does not inflate the estimate.
- **Materiality.** In the A–S model the reservation price is $r = s - q\gamma\sigma^2(T-t)$ and the optimal total spread is $\gamma\sigma^2(T-t) + \frac{2}{\gamma}\ln(1+\gamma/k)$. The measured variation in σ over a typical quoting horizon is fed through these formulas to obtain the implied quote error. The constant-volatility assumption is called materially violated if that error exceeds one tick. This threshold is fixed now, before the full analysis.

Q6: Return distribution. Heavy tails are a well-documented feature of asset returns (Cont, 2001); the question is their size at market-making horizons and for each coin.

- Excess kurtosis, QQ plots and the share of zero returns at 1 s, 10 s and 1 min.
- The Hill estimator (Hill, 1975) of the tail index, with a plot of the estimate against the number of tail observations used.
- To separate changing volatility from genuine jumps, the same analysis is repeated on returns divided by their GARCH volatility. If those standardized returns are close to normal, the tails come mainly from volatility changing over time; if not, there are jumps as well.

Q7: Fill intensity.

- **Empirical fill data.** At the start of each window of length τ, a hypothetical ask is placed δ ticks above the mid (and a bid δ ticks below). The outcome is 1 if any trade in the window reaches that price and 0 otherwise, for $\delta = 1, \ldots, 30$ ticks and $\tau \in \{1\text{ s}, 10\text{ s}, 60\text{ s}\}$.
- **Model.** Under A–S, fills arrive as a Poisson process with intensity $\lambda(\delta) = Ae^{-k\delta}$, so the chance of at least one fill in a window is $P(\delta) = 1 - \exp(-\tau Ae^{-k\delta})$. Rearranged,

$$\log\big(-\log(1-P)\big) = \log\tau + \log A - k\delta,$$

  which is exactly a **binomial generalized linear model with a complementary log-log link** (McCullagh and Nelder, 1989), an offset of $\log\tau$, and a linear term in δ whose coefficient is $-k$. The A–S assumption therefore becomes a standard GLM. Coin, side, volatility regime and hour enter as covariates and interactions, so k can differ across them.
- **Inference.** Outcomes within a window are nested (a hit at 10 ticks implies a hit at 5), so model-based standard errors are too small. Intervals come from the day-block bootstrap.

- **Limitation.** A price being reached is not the same as a resting order being filled, because the order may sit behind others in the queue. The fitted curve is an upper bound on true fill probability.

Q8: Adverse selection.

- **Markout curves.** For each parent trade at time t with sign s, the markout at horizon h is $M(h) = s\,(m_{t+h} - m_t)/m_t \times 10^4$ bps, for $h \in \{1, 2, 5, 10, 20, 30, 60\}$ seconds, with day-block bootstrap bands. A positive average means the price keeps moving the way the taker wanted, which is a loss for the maker.
- **Order-flow regression. OLS** of the next one-minute return on order-flow imbalance (buyer-initiated minus seller-initiated volume over total volume), quote imbalance and hour effects, by coin, with Newey–West errors, following the order-flow literature (Kyle, 1985; Hasbrouck, 1991; Cont et al., 2014).
- **Direction model. Logistic regression** of whether the next mid change is up or down on the same predictors. This gives an interpretable odds ratio (how much one-sided flow raises the odds of a move in its direction) and an AUC on March data.
- A small R^2 is expected. The consulting point is whether the effect is stable, because a small but reliable effect matters to someone quoting thousands of times a day.

3.4 Summary of questions, models and clients

Q	Topic	Main statistical models	Main client
1	What makers earn	Spread decomposition; linear mixed-effects; quantile regression	Desk
2	Fees	Logistic regression of profitable fills; break-even fee table	Desk, individual traders
3	Impact vs. size	Log-log regression, test of square-root slope	Desk, execution traders
4	Liquidity calendar	Linear mixed-effects; logistic regression; quantile regression	Execution traders, desk
5	Volatility	Hour-of-day regression; HAR-RV; GARCH and GJR-GARCH with t errors; Ljung–Box	Research labs
6	Return distribution	Student-t MLE vs. normal (AIC); Hill estimator; GARCH-standardized QQ	Research labs
7	Fill intensity	Binomial GLM, complementary log-log link; GAM spline with LR test; power law; logistic benchmark	Research labs, desk
8	Adverse selection	Markouts with block bootstrap; OLS with HAC errors; logistic regression	Research labs, desk

3.5 Tools

Python 3 with pandas, NumPy and SciPy; statsmodels for GLMs (including the complementary log-log link), GAMs, logistic, quantile and mixed-effects regression, VARs, Granger tests and HAC

errors; and DuckDB with Parquet for the full sample. All code is kept in a version-controlled repository with a pipeline that rebuilds every table and figure from the raw files.

4 Project Plan

4.1 Timeline

The plan runs from mid-October to the end of the semester. Dates are approximate and will be updated against the course calendar.

Dates	Work	Output
Oct 15–21	Scoping with clients; convert all 540 files to Parquet; integrity checks for every coin and day; verify row order; tick size and one-tick share per coin	Data-quality log; confirmed scope
Oct 22–Nov 1	Per-second, per-minute and parent-trade tables for all coins; trade–quote alignment; EDA on January–February	Sample EDA memo to clients (1–2 pages)
Nov 2–15	Part A core: Q1–Q2 (spread decomposition, fees); Q3–Q5 if on schedule	Draft results, Part A
Nov 16–25	Part B: Q6–Q7 (volatility, tails) and Q8 (fill model, adverse selection)	Draft results, Part B
Nov 26–29	Thanksgiving break; buffer	—
Nov 30–Dec 6	Evaluate all fitted models on March; robustness checks; stretch questions only if everything else is done	Results meeting with clients
Dec 7–end	Write final report and practitioner brief; prepare presentation	Final deliverables

Table 2: Project timeline.

4.2 Deliverables

1. **Final consulting report** (15–25 pages plus appendix), with a one-page non-technical executive summary that has a separate paragraph for each client. For research labs: which A–S assumptions to relax first and realistic parameter ranges by coin. For the desk: which coins are viable at which fee levels, how wide to quote, when to step back, and how to react to ETH moves. Followed by data, methods, findings with effect sizes, and limitations.
2. **Practitioner brief** (2 pages): a plain-language summary for traders, with the break-even fee table, the cost of trading different sizes, and the liquidity calendar for each coin.
3. **Presentation** to the class, aimed at a non-specialist audience.

References

Andersen, T. G., Bollerslev, T., Diebold, F. X., and Labys, P. (2000). Great realizations. *Risk*, 13:105–108.

Avellaneda, M. and Stoikov, S. (2008). High-frequency trading in a limit order book. *Quantitative Finance*, 8(3):217–224.

Binance (2024). Binance public data. https://data.binance.vision; documentation at https://github.com/binance/binance-public-data.

Cartea, Á., Jaimungal, S., and Penalva, J. (2015). *Algorithmic and High-Frequency Trading*. Cambridge University Press.

Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2):223–236.

Cont, R., Kukanov, A., and Stoikov, S. (2014). The price impact of order book events. *Journal of Financial Econometrics*, 12(1):47–88.

Granger, C. W. J. (1969). Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, 37(3):424–438.

Guéant, O., Lehalle, C.-A., and Fernandez-Tapia, J. (2013). Dealing with the inventory risk: a solution to the market making problem. *Mathematics and Financial Economics*, 7(4):477–507.

Hasbrouck, J. (1991). Measuring the information content of stock trades. *Journal of Finance*, 46(1):179–207.

Hasbrouck, J. (2007). *Empirical Market Microstructure*. Oxford University Press.

Hayashi, T. and Yoshida, N. (2005). On covariance estimation of non-synchronously observed diffusion processes. *Bernoulli*, 11(2):359–379.

Hill, B. M. (1975). A simple general approach to inference about the tail of a distribution. *Annals of Statistics*, 3(5):1163–1174.

Huang, R. D. and Stoll, H. R. (1996). Dealer versus auction markets: a paired comparison of execution costs on NASDAQ and the NYSE. *Journal of Financial Economics*, 41(3):313–357.

Koenker, R. and Bassett, G. (1978). Regression quantiles. *Econometrica*, 46(1):33–50.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6):1315–1335.

Laird, N. M. and Ware, J. H. (1982). Random-effects models for longitudinal data. *Biometrics*, 38(4):963–974.

McCullagh, P. and Nelder, J. A. (1989). *Generalized Linear Models*, 2nd edition. Chapman and Hall.

Sims, C. A. (1980). Macroeconomics and reality. *Econometrica*, 48(1):1–48.

Tóth, B., Lempérière, Y., Deremble, C., de Lataillade, J., Kockelkoren, J., and Bouchaud, J.-P. (2011). Anomalous price impact and the critical nature of liquidity in financial markets. *Physical Review X*, 1(2):021006.

Zhang, L., Mykland, P. A., and Aït-Sahalia, Y. (2005). A tale of two time scales: determining integrated volatility with noisy high-frequency data. *Journal of the American Statistical Association*, 100(472):1394–1411.

O'Connor, J., Abdullah, M., Umaima, S. M., and Parker, G. (2026). Quality-diversity for automated market-making under regime uncertainty. In *Proceedings of the 2026 IEEE Symposium on Computational Intelligence for Financial Engineering and Economics (CIFEr)*, Tokyo, Japan. IEEE.